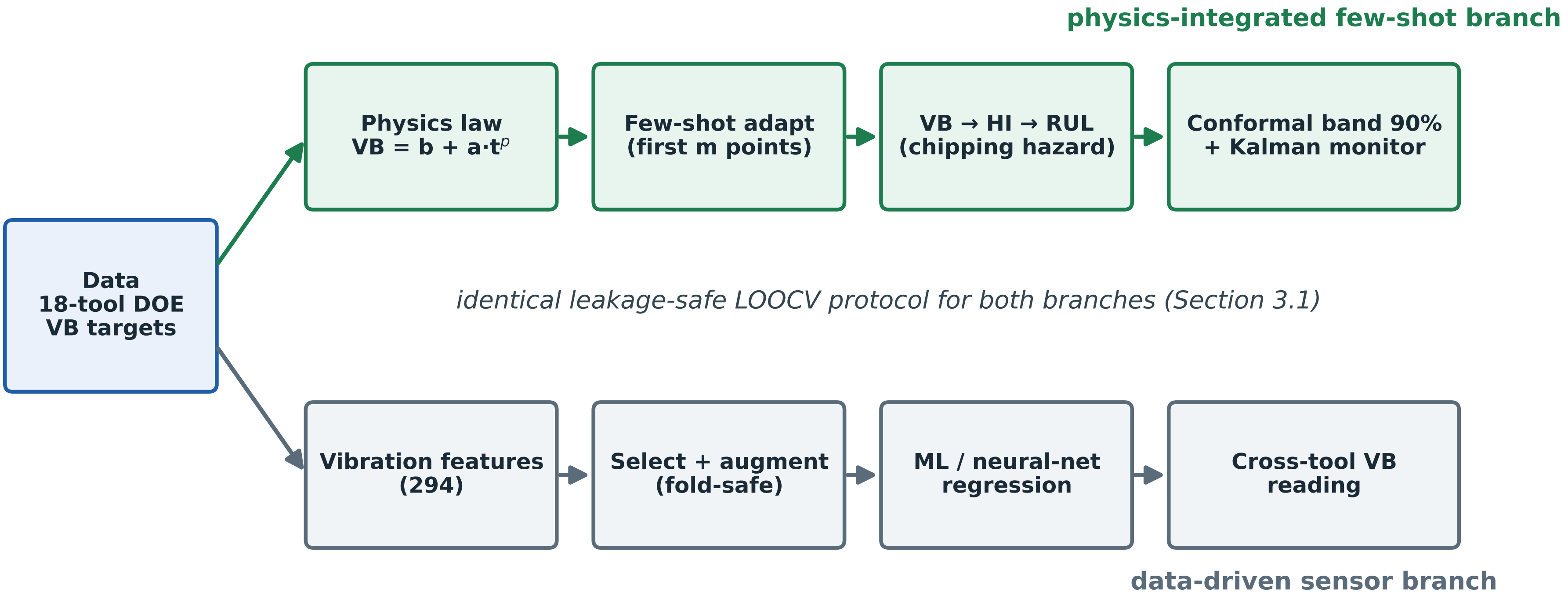
physics-integrated few-shot branch
Physics law
VB = b + a·t^p
Few-shot adapt
(first m points)
VB → HI → RUL
(chipping hazard)
Conformal band 90%
+ Kalman monitor
Data
18-tool DOE
VB targets
identical leakage-safe LOOCV protocol for both branches (Section 3.1)
Vibration features
(294)
Select + augment
(fold-safe)
ML / neural-net
regression
Cross-tool VB
reading
data-driven sensor branch